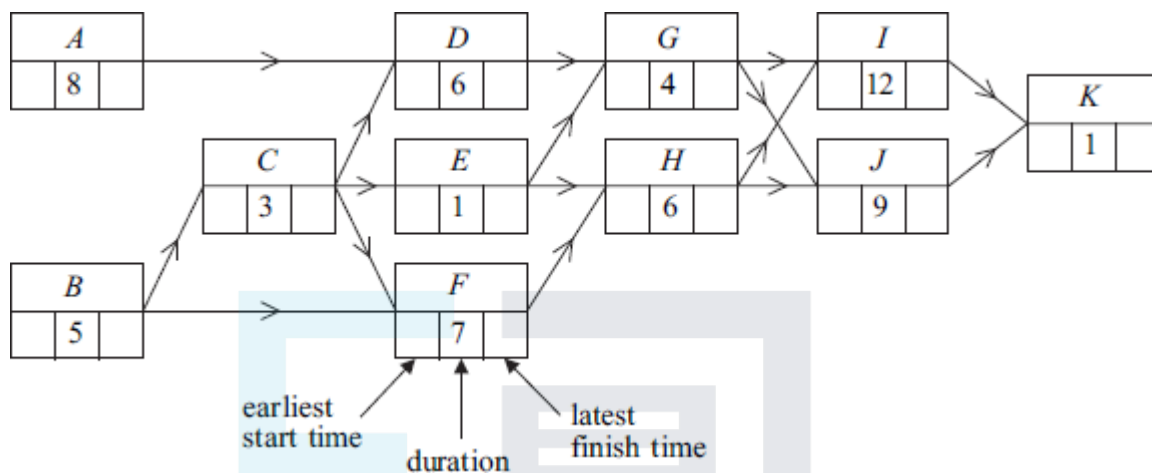


Q1.

Figure 1 below shows an activity diagram for a project. Each activity requires one worker. The duration required for each activity is given in hours.

- (a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

Figure 1



(4)

- (b) On **Figure 2** complete the precedence table.

Figure 2

Activity	Immediate predecessor(s)
A	
B	
C	
D	
E	
F	
G	
H	
I	
J	
K	

(2)

- (c) Find the critical path.

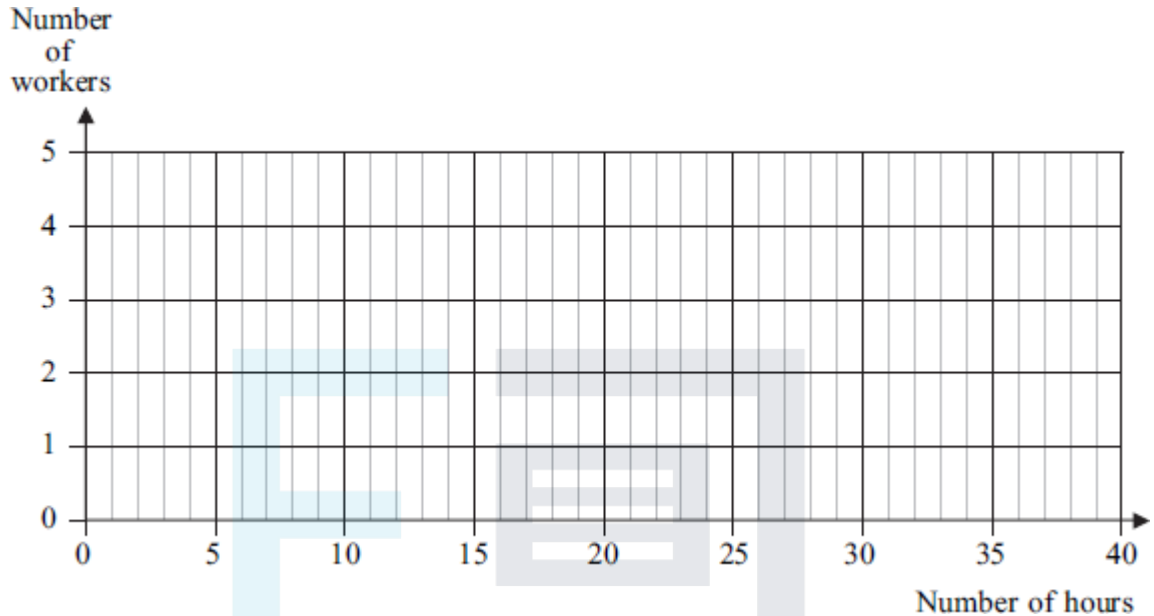
(1)

- (d) Find the float time of activity *E*.

(1)

- (e) Using **Figure 3**, draw a resource histogram to illustrate how the project can be completed in the minimum time, assuming that each activity is to start as early as possible.

Figure 3



(3)

- (f) Given that there are two workers available for the project, find the minimum completion time for the project.

(1)

- (g) Given that there is only one worker available for the project, find the minimum completion time for the project.

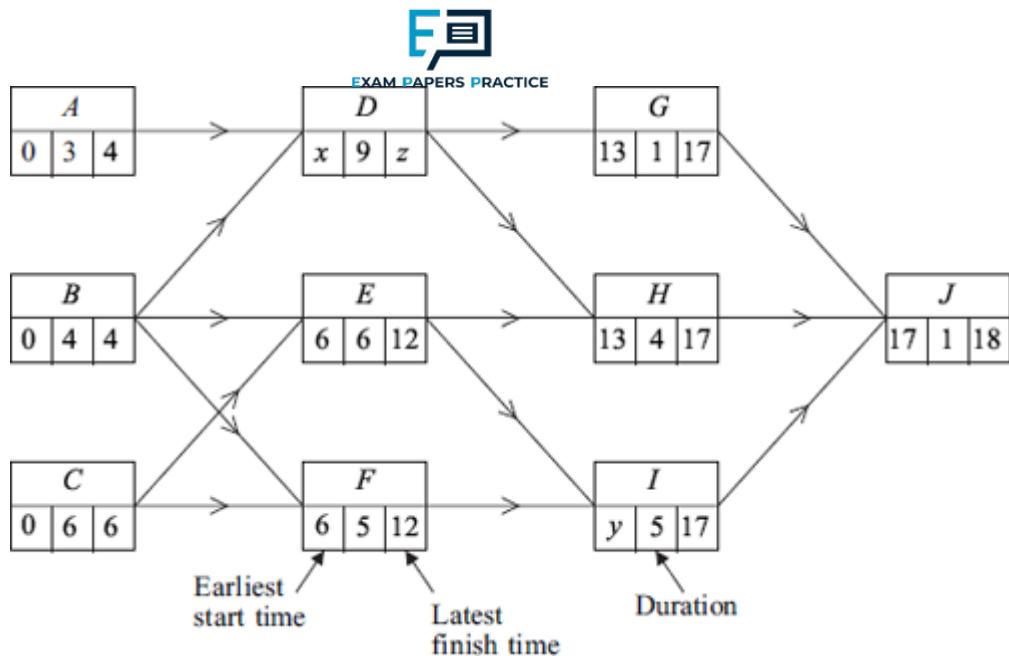
(1)

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(Total 13 marks)

Q2.

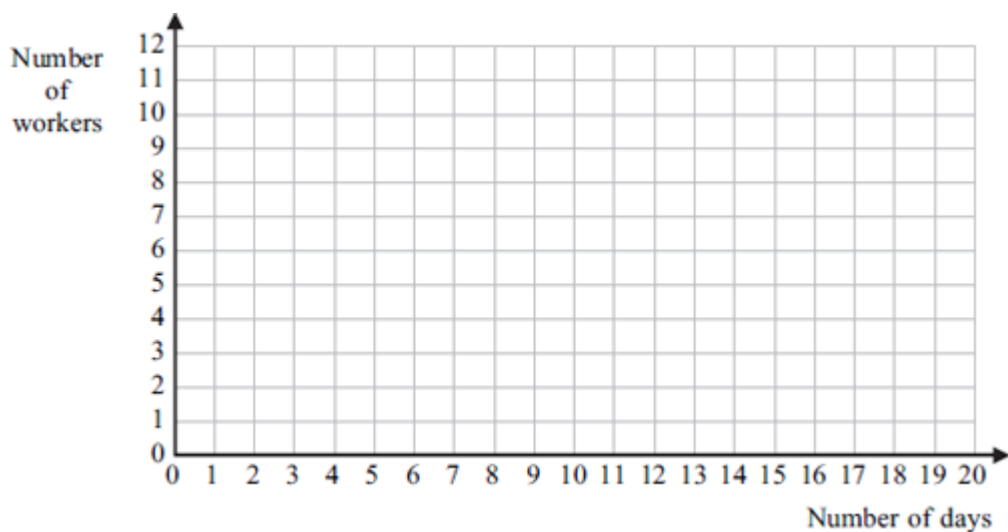
The diagram shows the activity network and the duration, in days, of each activity for a particular project. Some of the earliest start times and latest finish times are shown on the diagram.



- (a) Find the values of the constants x , y and z . (3)
- (b) Find the critical paths. (2)
- (c) Find the activity with the largest float and state the value of this float. (2)
- (d) The number of workers required for each activity is shown in the table.

Activity	A	B	C	D	E	F	G	H	I	J
Number of workers required	4	2	3	4	2	4	3	3	5	6

Given that each activity starts as **early** as possible and assuming that there is no limit to the number of workers available, draw a resource histogram for the project on the **figure** below, indicating clearly which activities are taking place at any given time.



- (e) It is later discovered that there are only 9 workers available at any time. Use resource levelling to find the new earliest start time for activity *J* so that the project can be completed with the minimum extra time. State the minimum extra time required.

(2)

(Total 14 marks)

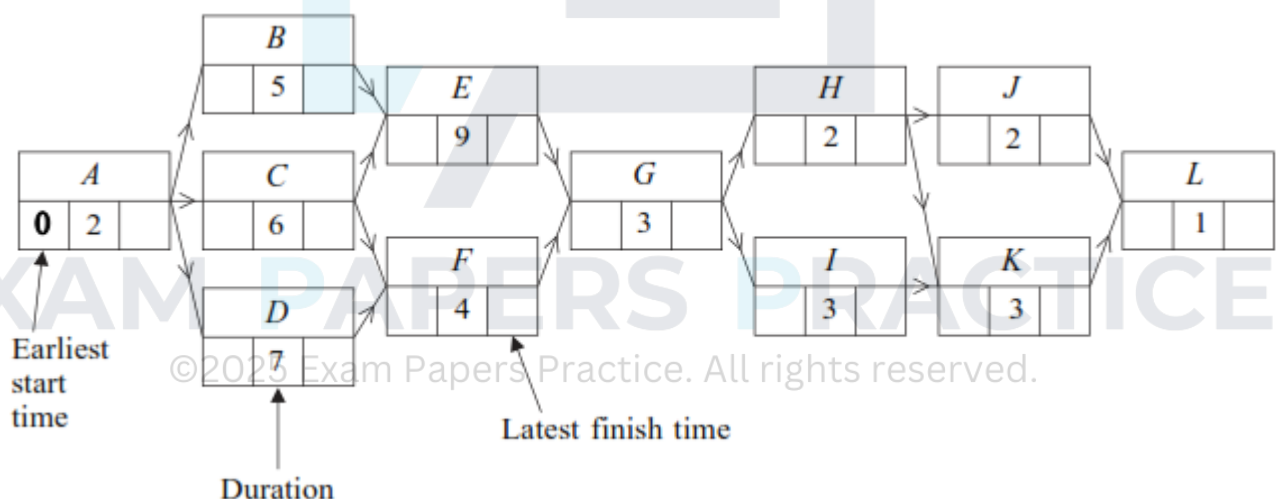
Q3.

A group of workers is involved in a decorating project. The table shows the activities involved. Each worker can perform any of the given activities.

Activity	A	B	C	D	E	F	G	H	I	J	K	L
Duration (days)	2	5	6	7	9	4	3	2	3	2	3	1
Number of workers required	6	3	5	2	5	2	4	4	5	3	2	4

The activity network for the project is given in **Figure 1** below.

Figure 1



- (a) Find the earliest start time and the latest finish time for each activity, inserting their values on **Figure 1**.

(4)

- (b) Hence find:

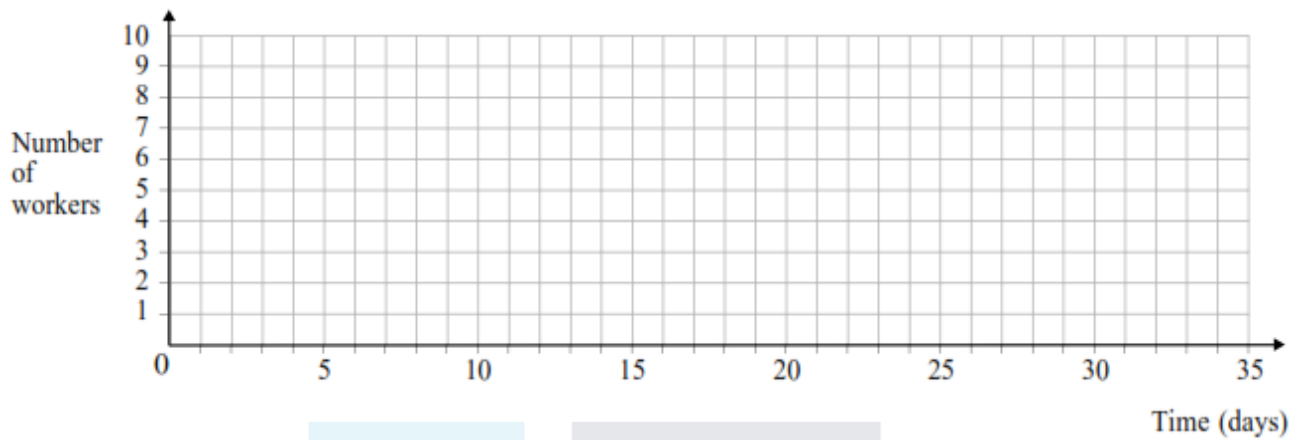
- the critical path;
- the float time for activity *D*.

(3)

- The critical path is _____
- The float time for activity *D* is _____

- (c) Given that each activity starts as early as possible and assuming that there is no limit to the number of workers available, draw a resource histogram for the project on **Figure 2** below, indicating clearly which activities are taking place at any given time.

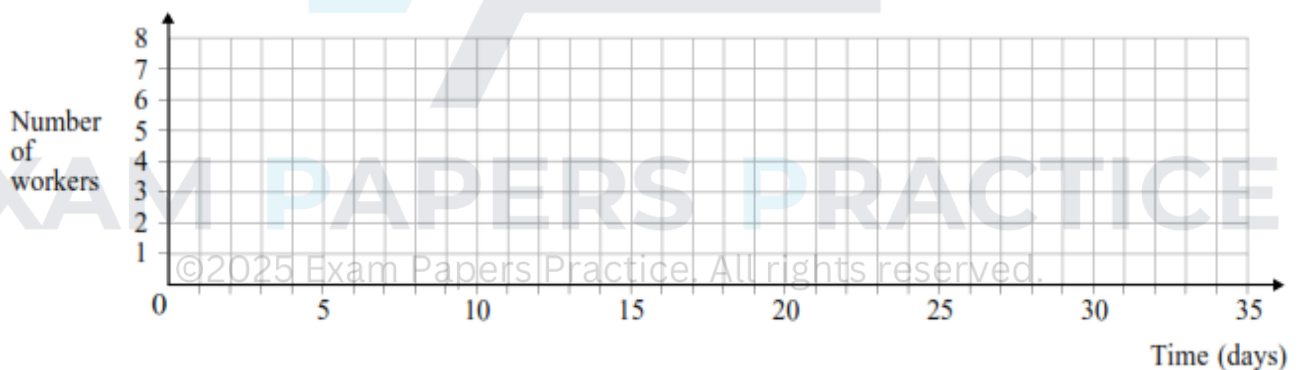
Figure 2



(4)

- (d) It is later discovered that there are only 8 workers available at any time. Use resource levelling to construct a new resource histogram on **Figure 3** below, showing how the project can be completed with the minimum extra time. State the minimum extra time required.

Figure 3



The minimum extra time required is _____

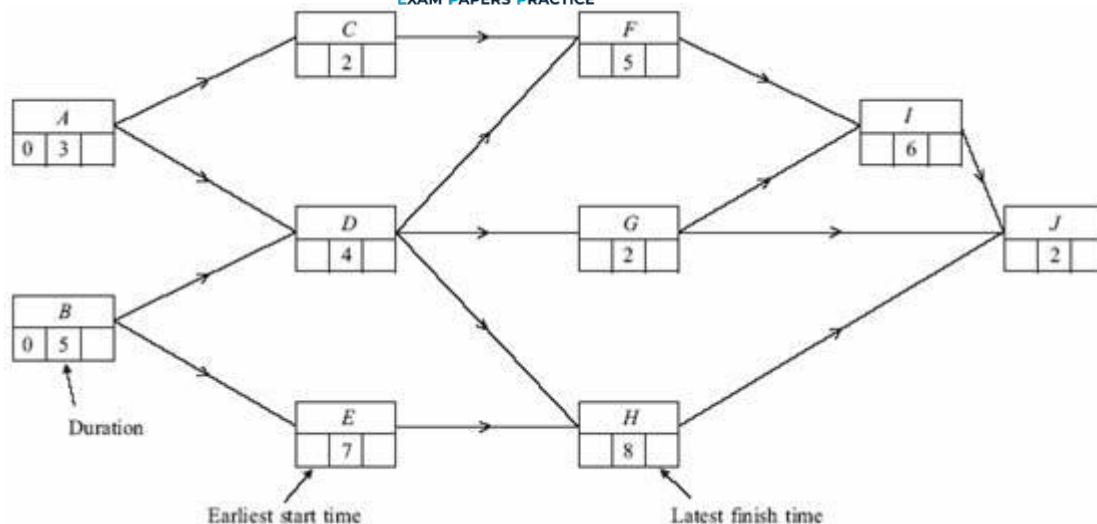
(3)

(Total 14 marks)

Q4.

Figure 1 shows the activity network and the duration in days of each activity for a particular project.

Figure 1

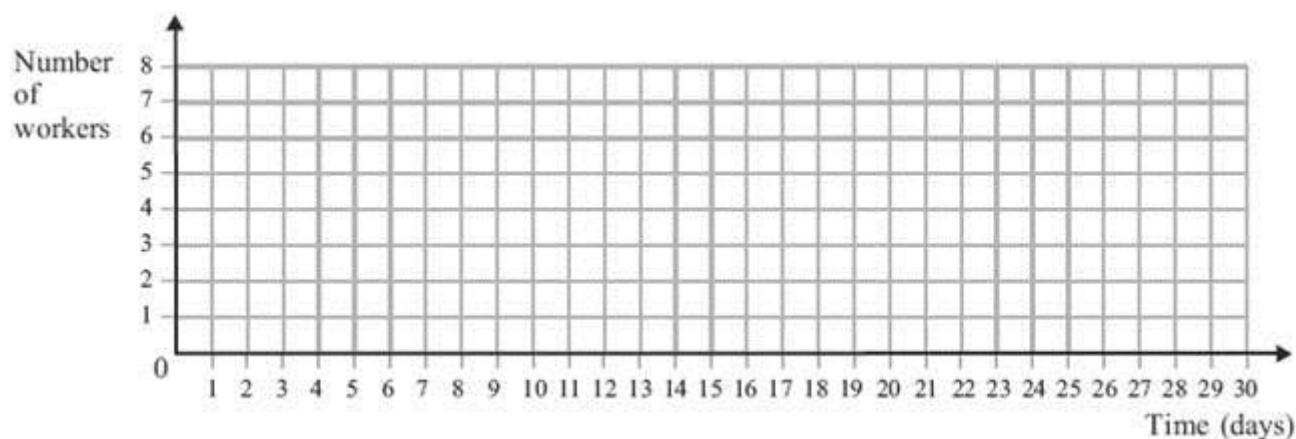


- (a) On **Figure 1**:
- find the earliest start time for each activity; (2)
 - find the latest finish time for each activity. (2)
- (b) Find the critical paths and state the minimum time for completion. (3)
- (c) The number of workers required for each activity is shown in the table.

Activity	A	B	C	D	E	F	G	H	I	J
Number of workers required	3	3	4	2	3	4	1	2	2	5

- (i) Given that each activity starts as early as possible and assuming that there is no limit to the number of workers available, draw a resource histogram for the project on **Figure 2**, indicating clearly which activities take place at any given time.

Figure 2



(4)

- (ii) It is later discovered that there are only 6 workers available at any time. Explain why the project will overrun, and use resource levelling to indicate which activities need to be delayed so that the project can be completed with the minimum extra time. State the minimum extra time required.

(3)

(Total 14 marks)

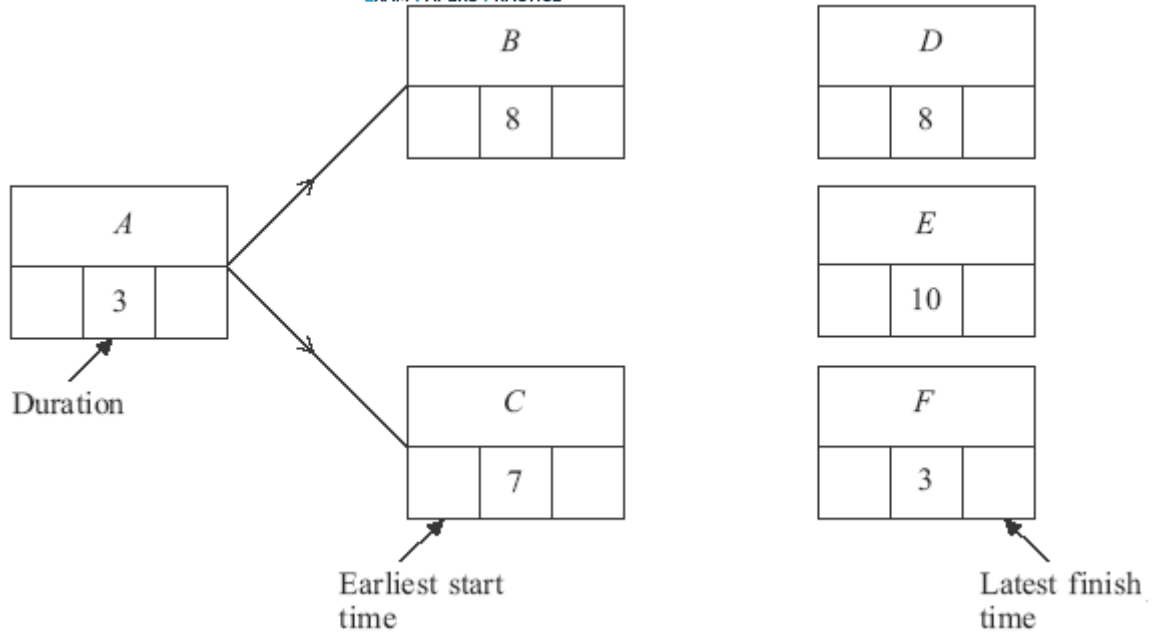
Q5.

A group of workers is involved in a building project. The table shows the activities involved. Each worker can perform any of the given activities.

Activity	Immediate predecessors	Duration (days)	Number of workers required
<i>A</i>	–	3	5
<i>B</i>	<i>A</i>	8	2
<i>C</i>	<i>A</i>	7	3
<i>D</i>	<i>B, C</i>	8	4
<i>E</i>	<i>C</i>	10	2
<i>F</i>	<i>C</i>	3	3
<i>G</i>	<i>D, E</i>	3	4
<i>H</i>	<i>F</i>	6	1
<i>I</i>	<i>G, H</i>	2	3

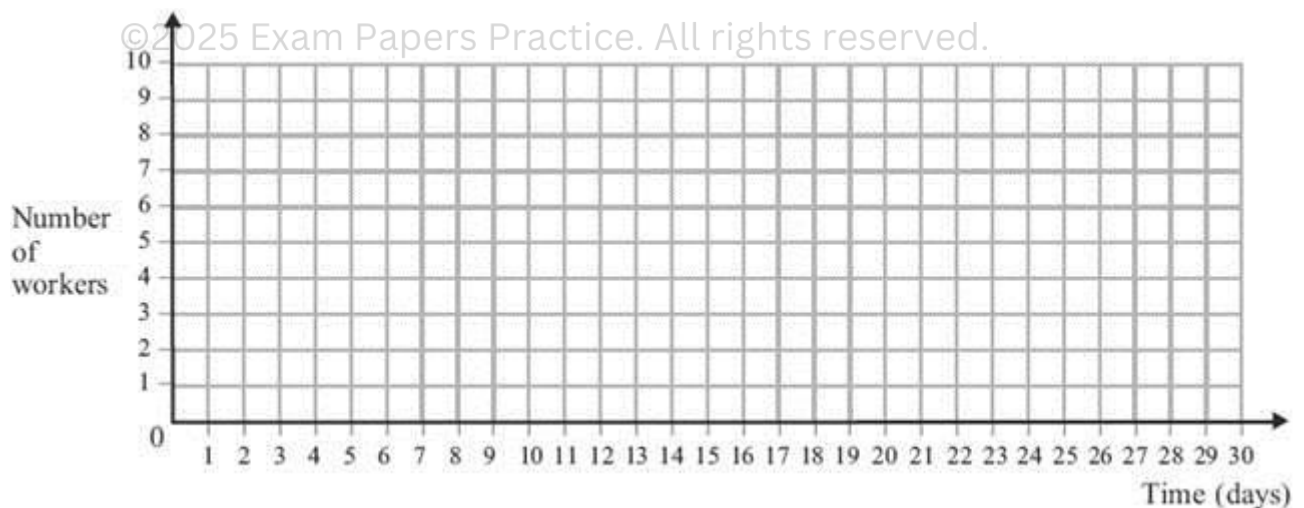
- (a) Complete the activity network for the project on **Figure 1**.

Figure 1



- (b) Find the earliest start time and the latest finish time for each activity, inserting their values on **Figure 1**. (2)
- (c) Find the critical path and state the minimum time for completion. (4)
- (d) The number of workers required for each activity is given in the table above. Given that each activity starts as early as possible and assuming there is no limit to the number of workers available, draw a resource histogram for the project on **Figure 2**, indicating clearly which activities take place at any given time. (2)

Figure 2



- (e) It is later discovered that there are only 7 workers available at any time. Use resource levelling to explain why the project will overrun and indicate which activities need to be delayed so that the project can be completed with the minimum extra (4)

time. State the minimum extra time required.

(3)
(Total 15 marks)

Q6.

A building project is to be undertaken. The table shows the activities involved.

Activity	Immediate Predecessors	Duration (days)	Number of Workers Required
<i>A</i>	–	2	3
<i>B</i>	<i>A</i>	4	2
<i>C</i>	<i>A</i>	6	1
<i>D</i>	<i>B, C</i>	8	3
<i>E</i>	<i>C</i>	3	2
<i>F</i>	<i>D</i>	2	2
<i>G</i>	<i>D, E</i>	4	2
<i>H</i>	<i>D, E</i>	6	1
<i>I</i>	<i>F, G, H</i>	2	3

(a) Complete the activity network for the project on **Figure 1**.

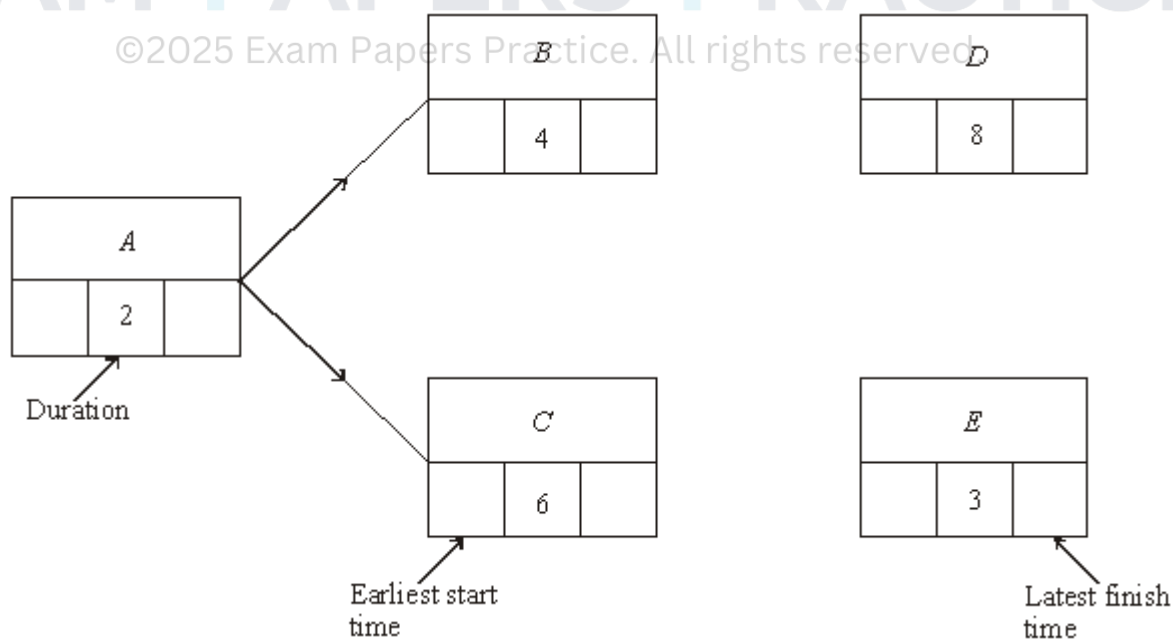


Figure 1

- (b) Find the earliest start time for each activity. (3)
- (c) Find the latest finish time for each activity. (2)
- (d) Find the critical path and state the minimum time for completion. (2)
- (e) State the float time for each non-critical activity. (2)
- (f) Given that each activity starts as early as possible, draw a resource histogram for the project on **Figure 2**. (2)

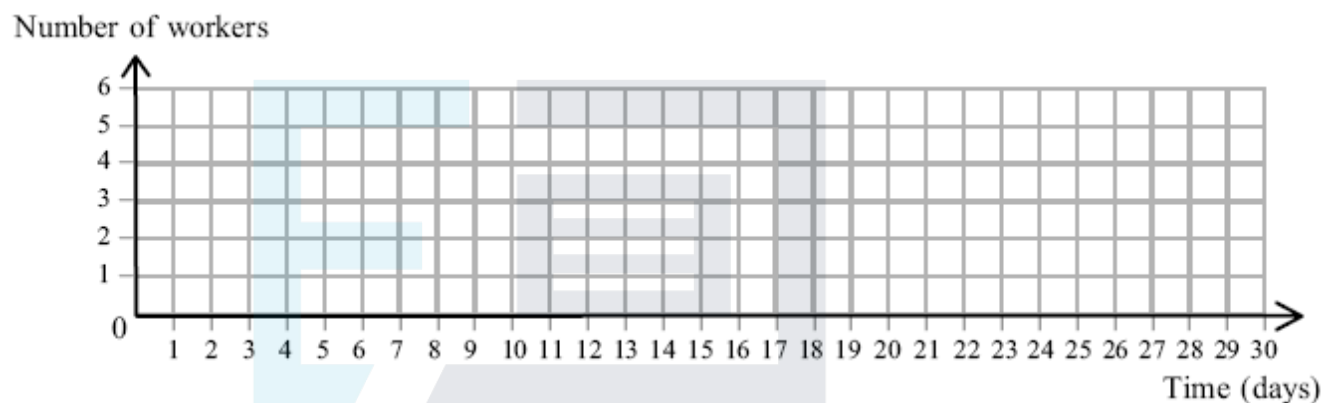
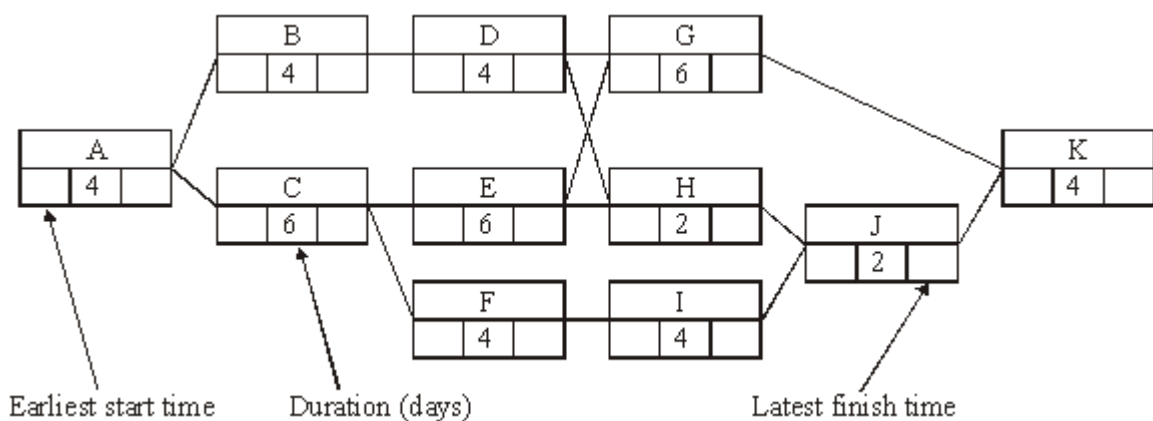


Figure 2

- (g) There are only 3 workers available at any time. Use resource levelling to explain why the project will overrun and state the minimum extra time required. (4)
- (3)
- (Total 18 marks)**

Q7.

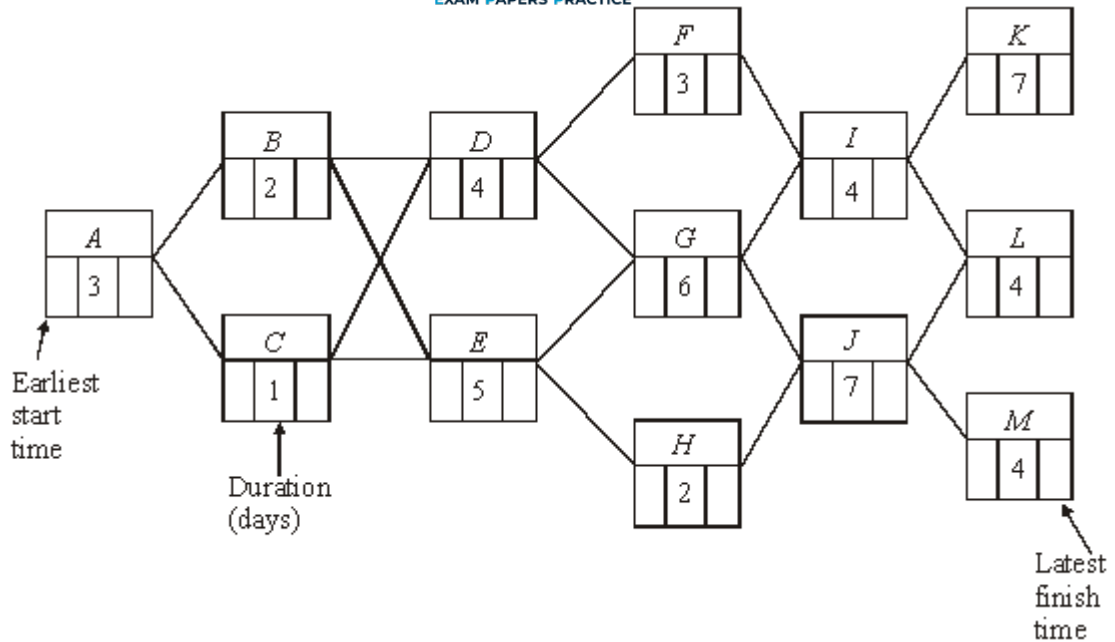
The diagram shows an activity network for a building project.



- (a) On the diagram:
- (i) find the earliest start time for each activity; (2)
 - (ii) find the latest finish time for each activity. (2)
- (b) Identify the critical path and state the minimum time for completion of the project. (1)
- (c) Given that one activity overruns by four days, but the completion time is unaffected, list the activities which could overrun. (1)
- (d) Each activity requires one worker in order for the activity to be completed in the stated time.
- Draw a resource histogram for the project. (3)
- (e) Given that there are only two workers available, each being able to complete any of the activities they undertake in the stated time:
- (i) state the new minimum time to complete the project, giving a reason for your answer; (2)
 - (ii) state how the activities could be divided between the two workers to achieve this time. (1)
- (Total 12 marks)**

Q8.

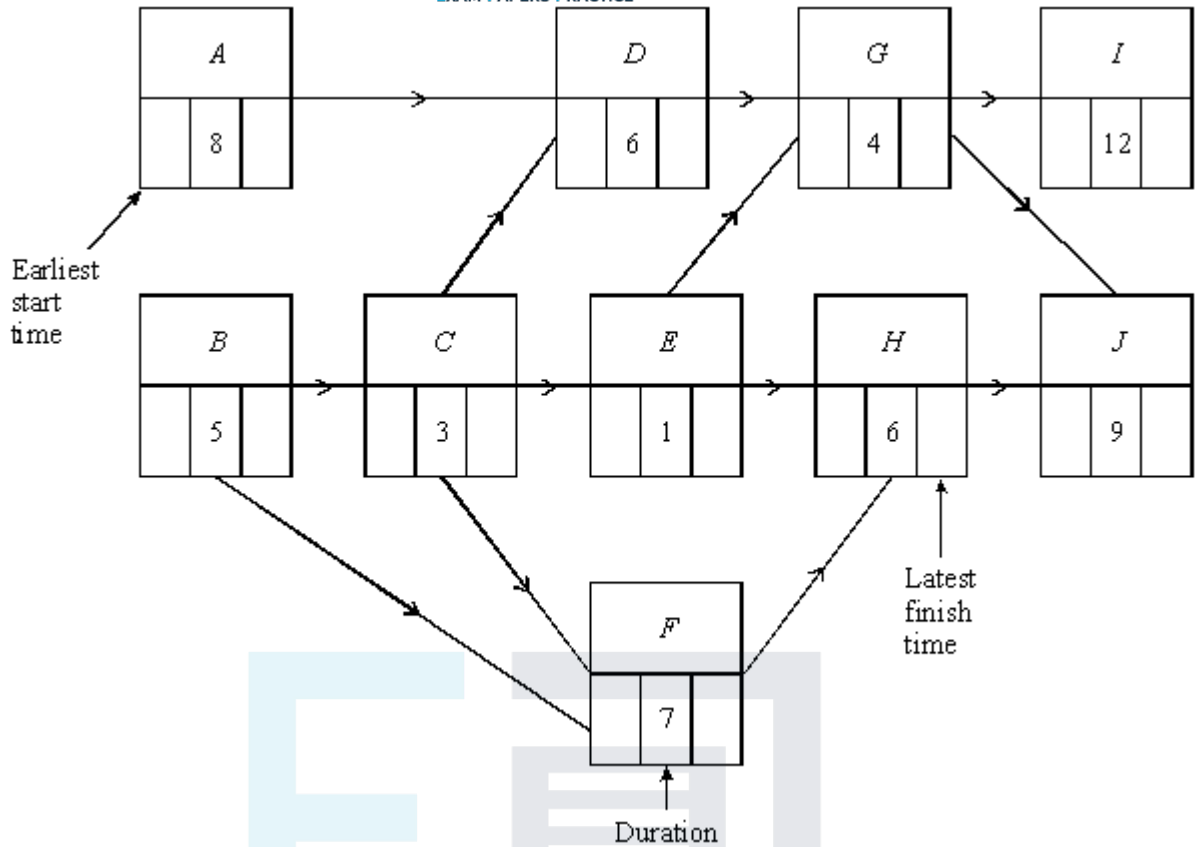
A major construction project is to be undertaken and the project has been divided into 13 activities. The following diagram shows the activity network for the project.



- (a) On the diagram above:
- find the earliest start time for each activity; (2)
 - find the latest finish time for each activity. (2)
- (b) List the non-critical activities. (1)
- (c) Draw a Gantt (cascade) diagram for the project. (3)
- (d) There are only two workers available, each able to complete any of the activities in the stated time.
- Find the new minimum completion time for the whole project. (2)
 - Allocate the activities to the two workers so that this new minimum time can be achieved. (2)
- (Total 12 marks)**

Q9.

The following diagram shows an activity network for a project. The time required for each activity is given in hours.



(a) On the table below, complete the precedence table.

Activity	Immediate Predecessor
A	
B	
C	
D	
E	
F	
G	
H	
I	
J	

(2)

(b) On the diagram above, complete the earliest start and latest finish times for each activity.

(4)

- (c) List the critical activities.

(1)

- (d) Draw a Gantt diagram, given that the project is to be completed in the minimum possible time and all the activities are to start as late as possible.

(4)

- (c) There are only two workers available. They are both capable of completing each activity on their own in the stated time.

Find the minimum overrun time for the whole project.

(2)

(Total 13 marks)

Q10.

A group of workers are involved in a self-build project. The project has been divided into the activities shown in the table below.

Activity	Immediate predecessors	Duration (days)	Number of workers required
A	—	2	8
B	A	6	4
C	A	12	4
D	B	12	8
E	B	16	4
F	B	4	6
G	C, F	10	6
H	E	4	4
I	D, G, H	2	8

- (a) Assuming that sufficient workers are available,

- (i) construct an activity network for the project,

(4)

- (ii) find the earliest start time for each activity,

(3)

- (iii) find the latest finish time for each activity,

(3)

- (iv) identify the critical path and state the shortest completion time for the whole project. (2)
- (b) Construct a resource histogram for the project showing the number of workers required each day, assuming each activity is to start as early as possible. (4)
- (c) Given that the group consists of 18 workers, explain why it would be necessary to reschedule some of the activities to enable them to finish the project in the shortest possible completion time. Give **two** possible ways in which this could be done. (4)
- (Total 20 marks)**

Q11.

A building project has been divided into a number of activities as shown in the table below.

	Activity	Immediate predecessor	Duration (days)
A	Prepare site	—	1
B	Erect walls	A	4
C	Excavate and lay drains	A	7
D	Fit window frames	B	3
E	Erect roof	B	9
F	Install electrics	B	7
G	Install plumbing	C, D	6
H	Insulate roof	E	3
I	Plaster walls	F, G, H	2

- (a) Construct an activity network for the project. (4)
- (b) Find the earliest start time for each activity. (3)
- (c) Find the latest finish time for each activity. (3)
- (d) Identify the critical path and state the shortest completion time for the whole project. (2)
- (e) Construct a cascade (Gantt) diagram for the project, assuming each activity is to

start as early as possible.

(4)

- (f) Each activity requires one worker. Draw a resource histogram showing the number of workers requires each day

(2)

- (g) Given that there are only three workers available on any one day, show how the activities could be allocated so that the project may be finished in the shortest completion time.

(2)

(Total 20 marks)

Q12.

[One sheet of 2 mm graph paper is provided for use in answering this question.]

Each year the secretary of a Mathematical Society undertakes the task of preparing and distributing the society's annual programme. The programme gives dates of meetings, a list of speakers and the titles of their talks, and an up-to-date list of paid-up members. The table below shows the activities which the secretary needs to carry out in preparing and distributing the programme.

	Activity	Immediate predecessor	Duration (days)
<i>A</i>	Select dates for meetings	—	3
<i>B</i>	Book speakers	<i>A</i>	13
<i>C</i>	Obtain advertising material for programme	<i>A</i>	12
<i>D</i>	Send out membership renewal notices	—	21
<i>E</i>	Prepare list of paid-up members	<i>D</i>	6
<i>F</i>	Prepare programme for printers and check proofs	<i>B, C, E</i>	10
<i>G</i>	Assemble material for circulation	<i>F</i>	7
<i>H</i>	Print members' address labels	<i>E</i>	4
<i>I</i>	Send out programmes	<i>G, H</i>	3

- (a) Construct an activity network for the project.

(4)

- (b) On your network show full details of the earliest and latest starting and finishing times for each activity. Indicate and state the critical path.

(6)

- (c) For each activity state the float time.

(2)

- (d) Construct a cascade diagram for the project.
If each activity requires one helper so that activities are completed on time, state the minimum number of helpers required for the project.

(5)

- (e) The secretary is let down by one helper who resigns immediately prior to the start of the project and so cannot be replaced. Find the least possible number of days' delay to the completion of the project and explain how it can be achieved.

(2)

(Total 19 marks)



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